IM0802 – Responsible Artificial Intelligence
Assignment 2: “Paper Title Goes Here”

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Abstract

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1 Introduction 350

Israel, which allocates 8.8% of its GDP to defense [1], invests heavily in the development of autonomous weapon systems (AWS), with their Prime Minister announcing a $100 billion investment in homegrown arms production last December [2]. Israel’s most well-known AWS is the Iron Dome air defense system, which intercepts incoming projectiles in real time. Building on this foundation in artificial intelligence (AI)-assisted defense technology, Israel has expanded into offensive targeting applications such as “Habsora”.

Habsora ("The Gospel") is an Israeli Defense Forces (IDF) AI-targeting system developed primarily by Unit 8200 within the Military Intelligence Directorate. First deployed during the May 2021 Operation "Guardian of the Walls", it generates target recommendations using a multimodal intelligence pipeline that encompasses signal intelligence, telephone metadata, satellite and drone imagery, social graph analysis, and life-pattern modelling [3, 4]. The system is not a single model, but an ensemble of multiple specialised algorithms fused together [4]. During the 2023-2024 Gaza conflict, Habsora was reported to generate over one hundred target recommendations per day, which in prior years required approximately twelve months of human analysts [3, 5].

Habsora’s AI capability is the probabilistic classification and prioritisation of locations and individuals as military objectives. It assigns confidence scores derived from large-scale pattern matching across a shared intelligence pool, rather than positive identification in the traditional intelligence analysis [6]. Military commands nominally authorise each recommendation, yet their review window documented for each target was approximately twenty seconds [3].

This creates a stakeholder dynamic with both ethical and legal significance. It’s output influencing both IDF commands, who operate under high cognitive load and time pressure; Palestinian civilians in the target environment bearing the lethal consequences and the broader international community which depends on International Humanitarian Law (IHL) compliance for the legitimacy of armed conflict. Steen et al. [7] offer a framework for this tension. They propose a two-axis model of Meaningful Human Control (MHC) which distinguishes between high automation and genuine human control, which both require adequate information and real freedom of action. The central argument of this report is that Habsora structurally undermines the second condition, and that this gap carries cascading implications across the technical, sociotechnical, and governance layers identified by Verdiesen et al. [8] in their Comprehensive Human Oversight Framework (CHOF).

2 Ethical Issue Identification 750

Verdiesen et al. [8]’s CHOF framework divides oversight into three layers, next to three phases. For each of these cells we capture the failures identified with Habsora’s implementation.

2.1 Technical Layer

Failures Deep neural network which inherently is opaque, can't see reasoning. No evidence that they have XAI tools like SHAP or Grad-CAM which can help.

Evidence Commanders report no visibility into which signals shape the decision. The IHL assessment requires to know *why* something was selected, else there would be no legal grounds.

Impact **?**] Maps into this with: **P-01** transparency, **P-03** accountability, **CH-06** no technical understanding & **CH09** no legal framework. Commanders cannot give real judgement as they cannot understand it with e.g. XAI.

2.2 Sociotechnical Layer

Failures Automation bias, people trust AI outputs too much, gradual stamp approvement. Also with the short time window this effect is amplified. Steen et al. [7] argue that you need two conditions to responsibility: adequate information and meaningful freedom of action. With the previous sentence discarding both. **?**] shows that from social science we have an explainer and explainee. Without showing what went into a decision people can't make real decisions when they only get a recommendation. Combined with the volume it makes it fully impossible, thus no moral or legal weight.

Evidence Former IDF commands mention rubber stamping/click approve.

Impact IDF commanders, get 'responsibility' for something they can't even evaluate + civilian targetet by the system. Which 'normally' takes a long time of real oversight and weighings, and are now reduced to a button press.

2.3 Governance Layer

Failures UNESCO says it's not legal to delegate life-or-death choices to an AI system. **?**]'s accountability cap is applicable. The responsibilities go: developers -> habsora algorithm -> IDF officer with recommendations -> approving commander. Thus spreading 'responsibility' when someone dies. No ethical/stakeholder impact assessment, such as VSD

Evidence 10% error tolerance in recommendations. Civilians are stakeholders, and certainly not consulted.

Impact World wide IHL

The CHOF structures oversight across three layers: governance, socio-technical layer, and technical. Each of these layers is further analysed across three temporal phases: before, during, and after deployment of the autonomous weapon system. The governance layer concerns the legal and ethical conditions, investigating whether societal values and ethical principles are respected. The socio-technical layer focuses on the stakeholders involved, both the ones operating the system and those personally affected by it. The technical layer looks at the system itself, addressing issues such as automation bias and the system working as a black box, meaning the inner workings of the system are not transparent to users or stakeholders.

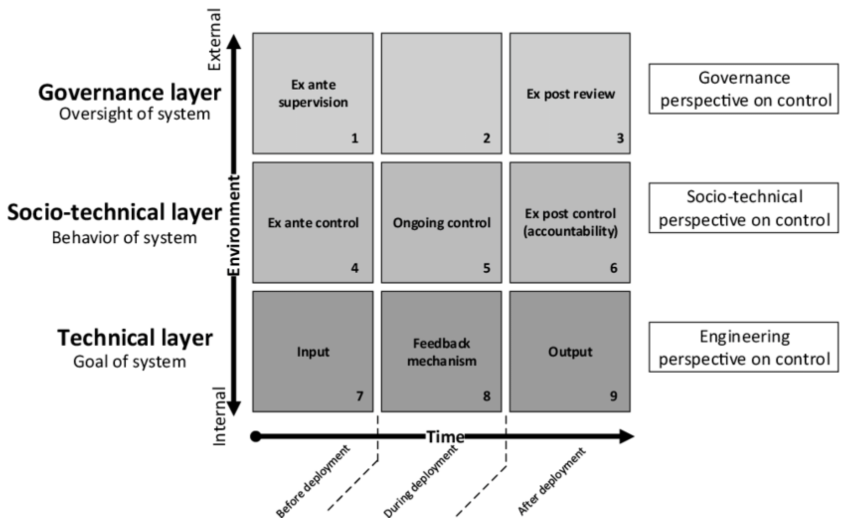

Figure 1: The CHOF framework for structuring ethical analysis across governance, socio-technical, and technical layers.

Underpinning all three layers is the Glass Box approach described by Aler Tubella et al. [9]. This approach translates abstract values, such as human dignity, into observable norms and requirements to be implemented in the the AI system. The Glass Box approach demands that the system's design and operation be transparent and verifiable, allowing for accountability and ensuring that ethical principles are upheld in practice. This is particularly relevant in the context of the IDF's use of the Lavender system, where the opacity of the AI's target recommendations raises significant ethical concerns regarding accountability and meaningful human control.

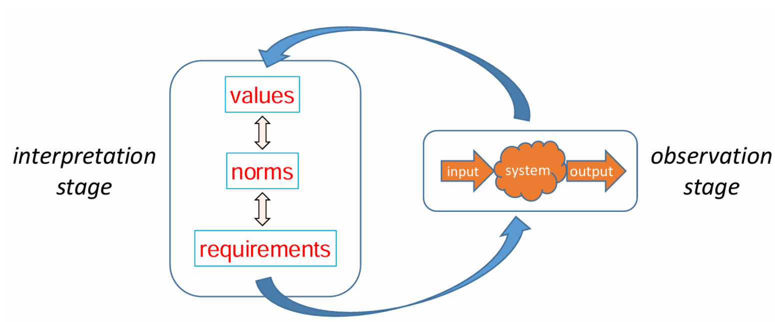

Figure 2: The two stages of the Glass-Box Approach: an Interpretation stage, where values are translated into design requirements, and an Observation stage, where we can observe and qualify the behaviour of the system

2.4 Governance Layer

Article 36 of 'Protocol I to the Geneva Conventions' requires countries to conduct legal reviews of new weapons International Committee of the Red Cross [10]. Specifically to determine whether its employment would, in some or all circumstances, be prohibited by international law. These legal reviews examine whether the system can distinguish civilians from combatants, whether it can be used in a way that limits unnecessary suffering, and whether meaningful human control is retained. This article, and the rest of Protocol I, apply to countries that have ratified (formally accepted) it. Israel is one of the few countries that have not [11] ratified Protocol I. This means that they are not legally obligated to conduct Article 36 reviews of new weapons systems. This forms a gap in governance because there is no legal requirement for Israel to assess the compliance of Habsora with international humanitarian law (IHL) before deployment.

Under International Humanitarian Law (IHL), human commanders remain legally responsible for how weapons are used, even if the system has autonomous functions [12]. In reality, commanders are judged on what they knew, or had reason to know, at the time of the decision, not on the basis of hindsight [13]. This creates accountability gaps when an AI system's recommendation leads to civilian casualties. Additionally, in a high-tempo conflict environment, where decisions must be made rapidly, the real-time use of the system can not be effectively monitored, further complicating accountability.

Post-deployment, there is often an absence of meaningful review processes to assess the system's compliance with IHL. It is difficult to attribute responsibility for any violations that may occur, as it may not be clear whether a particular recommendation was generated by the AI or if a human operator overrode it. This lack of transparency hinders accountability and undermines trust in the use of such systems in armed conflict.

2.5 Socio-technical Layer

It can be reasonably inferred that the affected stakeholders, Palestinian civilians, were likely not involved in the original design of Habsora. Their values were not elicited to ground the system's targeting logic. The values and needs of those most impacted by the system were not adequately considered.

IDF commanders operating under time pressure may be susceptible to automation bias, where they over-rely on the system's recommendations without sufficient critical evaluation [14]. This can lead to an erosion of genuine human control, as defined by Steen et al. [7]. The system's speed and opacity may make it difficult for operators to meaningfully review recommendations, further undermining human control over lethal decisions.

Palestinian civilians are non-consenting stakeholders who bear the consequences of the system's use without any recourse. There is a significant asymmetry between those operating the system and those affected by it, raising ethical concerns about justice and fairness in the deployment of such technologies in armed conflict.

2.6 Technical Layer

There were no verifiable, observable requirements anchoring the system's recommendations, and the underlying model was opaque to its operators [3]. This lack of transparency means that operators had no principled basis on which to contest or override a recommendation, as they could not inspect the criteria by which targets were selected.

The speed of output generation is incompatible with meaningful human review. Sources cited by +972 Magazine and Local Call report that human overseers devoted approximately 20 seconds to each target, limiting their verification to whether the target was male, before approving a strike [3]. This effectively reduces the operator to a rubber stamp [15], delegating life-and-death decisions to the system itself. When operators are structurally unable to critically evaluate recommendations due to high throughput constraints, the claim of human oversight becomes procedurally hollow.

The absence of audit trails makes post-hoc verification of norm compliance impossible. Without records of which recommendations were generated, which were overridden, and on what basis, it is impossible to reconstruct the decision-making process after the fact. This forecloses any meaningful accountability mechanism.

References

[1] Stockholm International Peace Research Institute. Trends in world military expenditure, 2024. SIPRI, April 2025. URL https://www.sipri.org/sites/default/files/2025-04/2504_fs_milex_2024.pdf. SIPRI Fact Sheet. Accessed: 07 April 2026.

[2] Times of Israel Staff. Israel investing more than $100 billion into independent arms industry, Netanyahu reveals. *The Times of Israel*, December 2025. URL https://www.timesofisrael.com/israel-investing-more-than-100-billion-into-independent-arms-industry-netanyahu-reveals/. Accessed: 12 April 2026.

[3] Yuval Abraham. “lavender”: the ai machine directing israel’s bombing spree in gaza, Apr 2024. URL https://www.972mag.com/lavender-ai-israeli-army-gaza/.

[4] Elizabeth Dwoskin. Israel built an “ai factory” for war. it unleashed it in gaza., Dec 2024. URL https://www.washingtonpost.com/technology/2024/12/29/ai-israel-war-gaza-idf/.

[5] Ynet. “man replaced by machine”: Is the use of ai undermining the idf’s intelligence capabilities?, Dec 2024. URL https://www.ynetnews.com/article/sjyhbn1ikx.

[6] Michael N. Schmitt and Jeffrey S. Thurnher. "out of the loop": Autonomous weapon systems and the law of armed conflict. *Harvard National Security Journal*, 4(2):231–281, 2013. URL https://centaur.reading.ac.uk/89863/.

[7] Marc Steen, Jurriaan Diggelen, Tjerk Timan, and Nanda Stap. Meaningful human control of drones: exploring human–machine teaming, informed by four different ethical perspectives. *AI and Ethics*, 3, 05 2022. doi: 10.1007/s43681-022-00168-2.

[8] Ilse Verdiesen, Andrea Aler Tubella, and Virginia Dignum. Integrating comprehensive human oversight in drone deployment: A conceptual framework applied to the case of military surveillance drones. *Information*, 12(9), 2021. ISSN 2078-2489. doi: 10.3390/info12090385. URL https://www.mdpi.com/2078-2489/12/9/385.

[9] Andrea Aler Tubella, Andreas Theodorou, Frank Dignum, and Virginia Dignum. Governance by glass-box: Implementing transparent moral bounds for ai behaviour. In *Proceedings of the Twenty-Eighth International Joint Conference on Artificial Intelligence, IJCAI-19*, pages 5787–5793. International Joint Conferences on Artificial Intelligence Organization, 7 2019. doi: 10.24963/ijcai.2019/802. URL https://doi.org/10.24963/ijcai.2019/802.

[10] International Committee of the Red Cross. Protocol additional to the geneva conventions of 12 august 1949, and relating to the protection of victims of international armed conflicts (protocol i), 8 june 1977, article 36 - new weapons. https://ihl-databases.icrc.org/en/ihl-treaties/api-1977/article-36, 1977. ICRC Database, Treaties, States Parties and Commentaries. Last accessed: 22 April 2026.

[11] International Committee of the Red Cross. State parties to Additional Protocol I of 1977. IHL Treaties Database, 2024. URL https://ihl-databases.icrc.org/en/ihl-treaties/api-1977/state-parties.

[12] International Committee of the Red Cross. Protocol I additional to the Geneva Conventions of 12 august 1949, article 86: Failure to act. IHL Treaties Database, 1977. URL https://ihl-databases.icrc.org/en/ihl-treaties/api-1977/article-86.

[13] Jean-Marie Henckaerts and Louise Doswald-Beck. *Customary International Humanitarian Law*, volume 1. Cambridge University Press, Cambridge, 2005.

[14] Raja Parasuraman and Dietrich H. Manzey. Complacency and bias in human use of automation: An attentional integration. *Human Factors: The Journal of the Human Factors and Ergonomics Society*, 52(3):381–410, 2010. doi: 10.1177/0018720810376055.

[15] Atay Kozlovski. When algorithms decide who is a target: Idf’s use of ai in gaza. *Tech Policy Press*, May 2024. URL https://www.techpolicy.press/when-algorithms-decide-who-is-a-target-idfs-use-of-ai-in-gaza/.